

1. Administrative & Study Identification

Full Study Title: A Phase 1, Open-label, Single-site, Multi-part, Non-randomized, Parallel-group Study to Assess the Relative Bioavailability of BMS-986435 Tablet Formulations (Parts 1 & 2) and Food Effect on the Selected BMS-986435 Tablet Formulation (Part 3) in Healthy Adult Participants
Short Title/Acronym: Bioavailability and Food Effect of BMS-986435 in Healthy Participants
Protocol Number: CV029-1024
NCT Number: NCT07226817
Sponsor Name: Bristol-Myers Squibb
Investigational Product: BMS-986435 (tablet formulations)
Phase of Development: Phase 1
Medical Monitor: BMS Clinical Trials Contact Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the relative bioavailability (rBA) of multiple test tablet formulations of BMS-986435 compared to an equal dose of the BMS-986435 reference tablet formulation.
Secondary Objectives: To evaluate the effect of food on the pharmacokinetics and absorption of the selected BMS-986435 tablet formulation, as well as to assess the general safety and tolerability of all formulations.

2.2 Background & Rationale BMS-986435 (also known as MYK 224 / Delocamten) is a cardiac myosin inhibitor being evaluated for the treatment of heart conditions, particularly Heart Failure with preserved Ejection Fraction (HFpEF) and Hypertrophic Cardiomyopathy. This Phase 1 study in healthy volunteers is conducted to assess the drug's absorption mechanisms across different tablet formulations. By determining relative bioavailability and assessing the impact of food on drug absorption, the study ensures optimized, consistent, and safe dosing regimens for future late-stage clinical trials.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional
Control Type: Active Comparator (Reference formulation vs. Test formulations)
Blinding: Open-label
Randomization: Non-randomized, Parallel-group (Parts 1 & 2), and Cross-over/Food Effect (Part 3)

3.2 Planned Enrollment & Duration Target **\$N\$:** Not publicly specified (Standard Phase 1 healthy volunteer cohort size)
Number of Sites: Single-site
Estimated Study Start: Late 2025
Estimated Primary Completion: TBD

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 65 years old. 2. Body weight of ≥ 45 kg and BMI between 18 and 32 kg/m^2 , inclusive. 3. Healthy as determined by medical history, physical examination, vital signs, 12-lead electrocardiogram (ECG), and clinical laboratory assessments. 4. Documented left ventricular ejection fraction (LVEF) $\geq 60\%$ and absence of significant cardiac abnormality at screening, as determined by local transthoracic echocardiogram (TTE) assessment.

4.2 Exclusion Criteria 1. Presence of any significant cardiac abnormality at screening. 2. Clinically significant abnormal laboratory values or vital signs indicative of underlying disease. 3. Other protocol-defined restrictions, including the use of prohibited concomitant medications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single or multiple doses of experimental BMS-986435 tablet treatments (evaluating experimental Treatments A through G). **Route of Administration:** Oral (PO, Tablet)
Duration of Treatment: Short-term exposure suitable for a Phase 1 bioavailability and food-effect study.

5.2 Concomitant & Prohibited Therapy Permitted: Standard permitted therapies per healthy volunteer Phase 1 guidelines.
Prohibited: Prescription and over-the-counter medications, specific supplements, and restricted foods (e.g., grapefruit, Seville oranges) that could alter the metabolism or absorption of the study drug.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Days -28 to -1	Informed Consent, Medical History, Physical Exam, ECG, TTE
Baseline	Day -1	Admission to clinical facility, baseline safety labs, vital signs
Interim	Dosing Days	Formulation Administration (Fasted/Fed), Serial PK Blood Draws, ECGs, Safety Monitoring
End of Study	Follow-up	Final Vital Signs, Exit Interview, Safety/Adverse Event Check

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Relative bioavailability (rBA) parameters of BMS-986435 test formulations compared to the reference formulation, measured by Maximum Observed Plasma Concentration (\$C_{max}\$) and Area Under the Plasma Concentration-Time Curve (\$AUC\$). **Measurement Method:** Serial plasma blood sampling analyzed via validated central bioanalytical laboratory methods.

7.2 Secondary & Exploratory Endpoints * Pharmacokinetic parameters (\$C_{max}\$, \$AUC\$) of the selected BMS-986435 tablet formulation administered under fed versus fasted conditions. * Incidence and severity of Treatment-Emergent Adverse Events (TEAEs), changes in clinical laboratory results, vital signs, and ECG parameters.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous inpatient monitoring during clinical confinement, physical exams, clinical laboratory testing, and ECG evaluations. **Serious Adverse Events (SAEs):** Standard expedited reporting to Sponsor/IRB within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety reviews conducted by the Principal Investigator and the Sponsor (Bristol-Myers Squibb).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: * **Pharmacokinetic (PK) Set:** All participants who receive at least one dose of the investigational product and have adequate concentration-time data. * **Safety Set:** All participants who receive at least one dose of the study drug. **Sample Size Justification:** Cohort sizes determined to provide adequate statistical power to estimate the 90% confidence intervals for the geometric mean ratios of PK parameters (\$C_{max}\$, \$AUC\$) between test and reference formulations. **Handling of Missing Data:** Standard Phase 1 protocols apply; missing PK timepoints are generally excluded from strict PK modeling but retained in safety evaluations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring by the sponsor to ensure protocol adherence, accurate data transcription, and participant safety. **Audit Trail:** Robust electronic Case Report Form (eCRF) tracking and data

clarification forms via standard clinical quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: CV029-1024 **Registry Number:** NCT07226817 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** Bioavailability and Food Effect of BMS-986435 **Official Regulatory Title:** A Phase 1, Open-label, Single-site, Multi-part, Non-randomized, Parallel-group Study to Assess the Relative Bioavailability of BMS-986435 Tablet Formulations (Parts 1 & 2) and Food Effect on the Selected BMS-986435 Tablet Formulation (Part 3) in Healthy Adult Participants

12. Clinical Framework (Pharmacokinetics / Phase 1 Specific)

Primary Condition: Healthy Adult Participants (Investigating therapy for Heart Failure/Cardiomyopathy) **Diagnosis Threshold:** BMI $\geq 18 - 32 \text{ kg/m}^2$; LVEF $\geq 60\%$ **Medical Methodology:** Relative Bioavailability & Food Effect Assessment **Optimization Target:** Identification of optimal oral formulation and dosing conditions (fed vs. fasted)

13. Study Procedures & Methodology

Management Pathway: Healthy Volunteer Clinical Pharmacology Pathway **Education Protocol (HCP):** Investigator protocol training and PK sampling standardization **Education Protocol (Patient):** Informed consent detailing dosing requirements, dietary restrictions, and clinical unit confinement rules **Quality Audit Mechanism:** Protocol deviation tracking, bioanalytical method validation, and clinical monitoring

14. Observation & Follow-Up Schedule

Total Duration: Short-term Phase 1 duration (Specific weeks dictated by the washout period between cross-over phases) **Onsite Visit Cadence:** Confinement during dosing and PK sampling, followed by designated outpatient safety visits **Remote Monitoring Frequency:** Standard safety check-ins post-discharge **Checklist Review Interval:** Daily during confinement phase

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |